CERTIFICATE OF CONFORMITY · BBICERT-2026-10-15

# Certificate of Systematic Verification and Conformity

Systematic verification of the repository big.bi-supplement at commit 4a0d76290aee, executed October 15, 2026. Item-by-item attestation on two axes: lineage identity to the 2023 engines and conformity to arXiv:2307.02299. 104 tests re-executed (104 passed; tally 92 CONFORM / 10 DOCUMENTED DEVIATION / 1 NON-CONFORM / 1 NOT APPLICABLE), nine-pair selector probe, equation identities at $\leq 4.441\times10^{-15}$, locus slopes within $4.6\times10^{-4}$, difference-5 exactly zero under the published conditions. Overall verdict: CONFORM WITH DOCUMENTED DEVIATIONS.

Attesting agent: Super Z (GLM, Z.ai)

Method: evidence rules S1–S10 and protocol P0–P6 of the agent brief of October 14, 2026 · standard [EXT]: comparative selection-vector audit, Oct. 13, 2026
Baselines consolidated: 104-test compliance audit · difference-5 report · differential supplement (Corrigenda 1–3)

# Certificate of Systematic Verification and Conformity

big.bi-supplement, commit `4a0d76290aee`

measured against arXiv:2307.02299 (Axis A) and the 2023 lineage (Axis L)

Attesting agent: Super Z (GLM, Z.ai) — automated verification run

October 15, 2026

| | |
|---|---|
| Overall verdict | **CONFORM WITH DOCUMENTED DEVIATIONS** |
| Tally (this run) | 104 executed, 104 passed — 92 CONFORM / 10 DOCUMENTED DEVIATION / 1 NON-CONFORM / 1 NOT APPLICABLE |
| Commit | `4a0d76290aee` (working tree content-identical; mode bits only) |
| Run duration | 602 s wall (104/104 passed; per-test statuses identical to the baseline, 104/104) |

**Abstract**

This certificate attests the result of a systematic verification of the repository `big.bi-supplement` at commit `4a0d76290aee`, executed on October 15, 2026 in an isolated environment (Python 3.12.14, numpy 2.5.3, scipy 1.18.1, matplotlib 3.11.2, pytest 9.1.1). Every item was measured on two distinct axes: **Axis L** (identity to the 2023 engines `Timit-to-Maeda` and `Syllable_Synthesis`, evidenced by the comparative note of October 13, 2026 [EXT]) and **Axis A** (conformity to the article arXiv:2307.02299). The verification re-executed the full 104-test compliance suite (all passed; compliance tally 92 CONFORM, 10 DOCUMENTED DEVIATION, 1 NON-CONFORM, 1 NOT APPLICABLE — per-test statuses identical to the October 2026 baseline), re-ran the nine-pair selector probe (partition 5 conform / 4 diverging, exactly as expected), and re-opened every `file:line` citation at the pinned commit. Equation identities hold at $\leq 4.441 \times 10^{-15}$ (tolerance $5 \times 10^{-15}$); locus slopes reproduce the article values within $4.6 \times 10^{-4}$ (tolerance $5 \times 10^{-4}$); the word-onset anchoring rule (difference 5) is exactly zero, bitwise, under all six published conditions. One item remains NON-CONFORM (the CO matrix Tip row, `constants.py:28`: $\Omega = 0.5$ undocumented), with bounded, quantified effect and a one-line correction path. The conformity statement covers the selection layer, the register constants, the trajectory mechanisms, the numerical reproduction of the article's equations and announced results, and the documentation state after Corrigenda 1–3; it does not extend to acoustic recordings, perceptual claims, or three-consonant clusters, for which the article defines no reference.

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# 1 Scope and sources

## 1.1 Parties of the verification

The verification was executed on October 15, 2026 by the attesting agent identified in §11, in the working directory of a local clone of the target repository. The parties and documents are those of the commissioning brief; they are restated here so that the certificate is self-contained.

Table 1: Parties and documents of the systematic verification.

| Role | Identification |
|---|---|
| Target repository | `big.bi-supplement`, commit `4a0d76290aee`, DOI 10.5281/zenodo.23239483 [1]; engine package `synthSYL/`, driver `vlam.py`, configuration `config.py`, demonstration scripts `scripts/`, manual `docs/manual.tex` |
| Reference article | arXiv:2307.02299 (Interspeech 2023), Eqs. (1)–(4), Table 1, §2.2, §3, §4 [2] |
| Standard (method) | External comparative note of October 13, 2026, 6 pp. [3], cited as [EXT] — document architecture and evidence rules S1–S10 |
| Lineage comparators | `Timit-to-Maeda`, `Syllable_Synthesis` (2023 engines) [4]; compared only through the excerpts of [EXT], not re-downloaded |
| Baseline audit | 104-test compliance audit and difference-5 report of October 2026 [5]; executable suite `tests/compliance/` (suites A–G) |
| Baseline documentation | Differential supplement [6] with Corrigenda 1–3 and the scope-of-impact rows 4a/4b |

## 1.2 Environment of the run

The run used the pinned audit virtual environment, identical to the baseline environment on every package. Imports of `synthSYL.phonology` and `synthSYL.constants` were confirmed before execution (`ART_CLUSTER` reads `[[0,1,2,5],[0,1,2]]` at import).

| | | | |
|---|---|---|---|
| Python 3.12.14 | numpy 2.5.3 | scipy 1.18.1 | matplotlib 3.11.2 |
| pytest 9.1.1 | commit `4a0d76290aee` | run date Oct. 15, 2026 | wall time 602 s |

One environment note is disclosed for transparency: a plugin closed matplotlib figures and forced garbage collection after each test. This is resource hygiene only — no test function, assertion, tolerance or recording path was modified; the per-test verdicts and statuses are byte-identical to the baseline JSON (§7). The suite was executed once, in a single session, and the JSON record was written by the session itself at the end of the run.

## 1.3 Pinning status

The verification is pinned to commit `4a0d76290aee`. The clone's `HEAD` is exactly this commit; the working tree differs from it only by file mode bits (`100644` → `100755` on 22 tracked files), with zero insertions and zero deletions of content — `git diff` over `synthSYL/`, `vlam.py` and `config.py` is empty of content change. The re-pinning rule of the brief therefore does not trigger: no engine file changed, and all verdicts are issued at the pinned hash. The repository was not modified outside the certificate's own source files: the engine, the drivers and the configuration are untouched (§10, step 10).

### 1.4 The two comparison axes

Every item of this certificate is measured on two distinct axes, and the summary table of §9 carries an explicit Axis column. **Axis L (lineage identity)** asks whether the 2026 repository implements the same selection policy as the two 2023 engines; its evidence is [EXT] plus the chain of custody (byte-for-byte match of `ART_CLUSTER` at `constants.py:89-92` and of the threshold rule at `phonology.py:337-341`, re-verified in this run). **Axis A (article conformity)** asks whether the repository conforms to the article [2]; its evidence is the executed suite and the nine-pair probe of Annex C. The two axes answer different questions and can disagree: on the cluster rule the repository is identical to its lineage (Axis L: **CONFORM**) while diverging from the article on four of nine pairs (Axis A: **DOCUMENTED DEVIATION**). Conflating the axes would produce exactly the confusion the standard exists to prevent.

### 1.5 Conventions and conversions

Parameter indices are 0-based Maeda indices: 0 Jaw, 1 Body, 2 Dorsum, 3 Tip, 4 Lip protrusion, 5 Lip height, 6 Hyoid [EXT §1]. Three conversions are used in this certificate and shown at each point of use (rule S2): (i) article numbering = code index +1, so code `[0,1,2,5]` = $\{1, 2, 3, 6\}$ in article numbering; (ii) the 1-based MATLAB-style tables of `Syllable_Synthesis` convert by $-1$ with zero-padding dropped and duplicates treated as inert (rule S3); (iii) angular quantities are compared modulo $2\pi$ (rule S9), so $-4.5\pi/8 \equiv 23\pi/16$ and $-\pi/12 \equiv 23\pi/12$. Place coding for the cluster rule is b= 0, d= 1, g= 2 (`_TABCONS_IDX`, `phonology.py:133-141`, quoted verbatim in Annex A).

## 2 Isolated consonants (item I1)

### 2.1 Requirement and evidence

The article's §3 defines the isolated-consonant tuning quoted here verbatim (extraction of October 15, 2026, from the article PDF): "...For the /g/ this is well known that it is velar for back vowels and palatal for front vowels. This led to the locations ($\rho_{gv} = 1.2,\ \theta_{gv} = \pi/3$) and ($\rho_{gp} = 1.1,\ \theta_{gp} = 23\pi/12$) combined with the selection vector $S_c = \{1, 2, 3, 4\}$ involving tongue and jaw. The location of /d/ depends on the following vowel around ($\rho_d = 1.2,\ \theta_d = 3\pi/2$) and /d/ has same selection vector $S_c = \{1, 2, 3, 4\}$." The /b/ is placed as the /u/ vowel with vowel-side selection $S_v = \{3, 4, 5, 7\}$; the engine's consonant-side selector for /b/ is $\{1, 2, 6\}$ in article numbering (jaw, body, hyoid), matching the repository's documented reading of the article's §3.

The repository's isolated selectors are read from `phonology.py:173-180` at the pinned commit and converted by +1 (rule S2) at the point of use:

| Consonant | Code selector (`phonology.py:173-180`) | +1 conversion | Article $S_c$ | Verdict (A and L) |
|---|---|---|---|---|
| /b/ | `[0, 1, 5]` | $\{1, 2, 6\}$ | $\{1, 2, 6\}$ | **CONFORM** |
| /d/ | `[0, 1, 2, 3]` | $\{1, 2, 3, 4\}$ | $\{1, 2, 3, 4\}$ | **CONFORM** |
| /g/ | `[0, 1, 2, 3]` | $\{1, 2, 3, 4\}$ | $\{1, 2, 3, 4\}$ | **CONFORM** |

### 2.2 Executed evidence and verdict

Three layers of independent evidence support item I1 (independence rule of the brief: executed check *and* repository citation). First, the executed suite: `test_C1_selection_vector_per_`

`consonant[b]`, `[d]`, `[g]` and `test_C2_partition_sv_plus_sc[b,d,g]` all returned CONFORME in this run (fresh `test_results.json`, October 15, 2026). Second, the repository citation `phonology.py:173-180` quoted above, with the polar values carried by the same entries ($\rho = 1.2$ for /d/; $\theta = 23\pi/16$ for /d/ — see §5 for the $\pi/16$ analysis). Third, [EXT §2.3]: "The isolated selectors are identical after index conversion." The singleton fallback for coronal/dorsal places (`phonology.py:303-304`, `return [0, 1, 2, 3]`) agrees with the same article vectors.

**Residual statement — isolated consonants**

Verified: the three isolated selectors /b/, /d/, /g/ on both axes, with executed and quotable evidence. Not covered here: the polar targets attached to the entries (item I6, §5) and the vowel-side selection $S_v$, which the article defines only through the /u/-placement argument; $S_v$ was exercised indirectly by the end-to-end tests of §7 but is not the subject of a separate item.

# 3 Binary clusters (items I2–I3)

## 3.1 Masks and rule, quoted at the pinned commit

The cluster masks are defined in `constants.py:89-92` and the selection rule in `phonology.py:337-341`; both are quoted verbatim (re-opened October 15, 2026):

```
ART_CLUSTER: list[list[int]] = [
  [0, 1, 2, 5], # labial-dental cluster
  [0, 1, 2], # dental-dorsal cluster
]

if m == 2:
  idx_c1 = _TABCONS_IDX.get(synt_keys[0], 0)
  idx_c2 = _TABCONS_IDX.get(synt_keys[1], 0)
  art_idx = int(idx_c1 + idx_c2 > 2)
  return ART_CLUSTER[art_idx]
```

The place coding comes from `_TABCONS_IDX` (`phonology.py:133-141`): `"b": 0, "p": 0, "d": 1, "t": 1, "g": 2, "k": 2`. The decision rule is thus art_idx $= \mathbf{1}[\text{idx}_{c1} + \text{idx}_{c2} > 2]$ — the same rule the standard found in both 2023 engines: "The decision rule is the same" [EXT §3.2], with the partition {bb, bd, db, bg, gb, dd} (sum 0–2, mask `[0,1,2,5]`) versus {dg, gd, gg} (sum 3–4, mask `[0,1,2]`). The code vector `[0,1,2,3]` does not exist in `ART_CLUSTER`; no re-execution is needed to see this — the two-line literal above is the entire mask table — and the probe of Annex C confirms it behaviorally.

## 3.2 Nine-pair probe (executed this run)

The probe of Annex C was executed on October 15, 2026 with the pinned repository on `sys.path`; its raw output is quoted in Annex C and summarized here with the place-index sums (b= 0, d= 1, g= 2; threshold: sum $> 2$):

| Pair | Place sum | Engine (article numbering, +1) | Article rule | Verdict (Axis A) |
|---|---|---|---|---|
| /bb/ | 0 | $\{1,2,3,6\}$ | $\{1,2,3,6\}$ | **CONFORM** |
| /bd/ | 1 | $\{1,2,3,6\}$ | $\{1,2,3,6\}$ | **CONFORM** |
| /bg/ | 2 | $\{1,2,3,6\}$ | $\{1,2,3,6\}$ | **CONFORM** |
| /db/ | 1 | $\{1,2,3,6\}$ | $\{1,2,3,6\}$ | **CONFORM** |
| /gb/ | 2 | $\{1,2,3,6\}$ | $\{1,2,3,6\}$ | **CONFORM** |
| /dd/ | 2 | $\{1,2,3,6\}$ | $\{1,2,3,4\}$ | **DOCUMENTED DEVIATION** |
| /dg/ | 3 | $\{1,2,3\}$ | $\{1,2,3,4\}$ | **DOCUMENTED DEVIATION** |
| /gd/ | 3 | $\{1,2,3\}$ | $\{1,2,3,4\}$ | **DOCUMENTED DEVIATION** |
| /gg/ | 4 | $\{1,2,3\}$ | $\{1,2,3,4\}$ | **DOCUMENTED DEVIATION** |

The partition is exactly the expected one: five conforming pairs (all containing /b/) and four diverging pairs (dd, dg, gd, gg). The divergence is ancestral, not a 2026 innovation: the standard established that the same masks and the same threshold rule are present in both 2023 engines [EXT §3.1–§3.2], so the 2026 repository did not "implement the article rule differently" — no engine of the lineage ever implemented that rule at all. The article's /b/-conditional rule ($S_c = \{1,2,3,6\}$ when /b/ is involved, otherwise $\{1,2,3,4\}$; "When involved, the /g/ is systematically palatal") is faithfully transcribed by the repository's manual but was never the engine's behavior.

### 3.3 Verdicts

**Axis L (I2, I3): CONFORM.** The masks and the threshold rule are byte-for-byte those of the lineage (chain of custody re-verified at `constants.py:89-92` and `phonology.py:337-341`; [EXT §3.1: "The masks are therefore identical"; §3.2: "The decision rule is the same"). **Axis A (I2): DOCUMENTED DEVIATION.** Four of nine pairs diverge from the article's /b/-conditional rule; the difference is documented repository-side by the differential supplement (row 4a and Corrigendum 2) [6], and its scope is bounded by §8 below (no article-cited result affected). The executed suite agrees per pair: `test_C4_cluster_selection_vectors[pair0-pair4]` CONFORME, `[pair5-pair8]` ÉCART DOCUMENTÉ (fresh `test_results.json`, this run), and the /gb/ cluster — the only one the article uses — is validated end-to-end at $\leq 2.220 \times 10^{-15}$ (`test_C4b`, §7).

## 4 Cluster activation and the $m = 3$ branch (item I4)

### 4.1 Trigger logic

Cluster mode is triggered by the syllabic parser: `cluster_art_params` is called with two (`m == 2`) or three (`m == 3`) consonant keys (`phonology.py:336-352`). The $m = 2$ branch is the threshold rule quoted in §3. The $m = 3$ branch exists in the target and is quoted here verbatim (re-opened October 15, 2026; this citation extends the baseline register):

```
if m == 3:
  places = set(_TABCONS_IDX.get(k, 0) for k in synt_keys if k)
  has_labial = 0 in places ; has_dorsal = 2 in places
  has_uvular = "R" in cons_keys[1]
  params:  set[int] = {0, 1, 2}
  if has_labial or not has_dorsal:  params.add(5)
  if has_uvular:  params.add(6)
```

[1]The repository string uses the IPA uvular turned-r glyph (U+0281); it is rendered as ASCII `R` here, exactly as the standard itself renders that glyph [EXT]. The quotation is otherwise verbatim.

```
    return sorted(params)
```

### 4.2 Standing against the standard and the article

The standard found three-consonant support only in `Timit-to-Maeda` and listed the $m = 3$ masks among its residual differences [EXT §4.3, §7]; the target's branch, read above, is the same generic construction (base $\{0, 1, 2\}$ plus labial/uvular extensions). On Axis L the item is therefore **CONFORM**(branch present, construction of the same family as [EXT §4.3]). On Axis A the item is **NOT APPLICABLE**: the article's CC text is descriptive only and defines no reference for CCC clusters, and the executed suite records exactly that status (`test_C5_triple_cluster`, NOT APPLICABLE, note: "extension du code au-delà de l'article"). The branch was nevertheless exercised by the executed suite (`test_E7_multiple_consonant_groups`, CONFORME this run), so its presence is not merely textual.

**Residual statement — clusters and activation**

Verified: the two cluster masks, the threshold rule, the nine-pair partition, the trigger logic for $m = 2$ and the presence and construction of the $m = 3$ branch. Left out, per the standard's own scope discipline [EXT §7–§9]: acoustic identity of CCC outputs (no article reference exists) and the polar targets of cluster consonants (item I6). A reader claiming acoustic identity for three-consonant sequences must check them separately.

## 5 Register constants and polar targets (items I5–I6)

### 5.1 The register family, quoted and adjudicated

The register constants are the numeric core of the engine. Each row of Table 2 carries the verbatim value at the pinned commit, the article reference where one exists, and the documentation adjudication. The CO matrix rows are read from `constants.py:24-32` (row 3 = Tip at `:28`); the dental angle from `:46`; the radii from `:74-81`; the VLAM soft-rectification ruling from `config.py:43-46`. All citations were re-opened on October 15, 2026.

**The Tip record (the single NON-CONFORM).** The manual claims the CO matrix "matches the article's Table 1 almost exactly; the only difference is the Tip parameter $c_0$: $-2.75$ vs $-3.0$" (`manual.tex:225-227`). That claim is accurate for $c_0$ but incomplete: the same row carries $\Omega = 0.5$ where the article has 0, and no repository document records this second difference. The baseline audit's counterfactual is decisive and was re-confirmed by this run's `test_A5_tip_coefficient_sensitivity` (ÉCART DOCUMENTÉ, executed): substituting the article's Tip row restores identity with the article's Table 1 in all seven coefficient checks at $\leq 1 \times 10^{-15}$, and the engine-level effect is confined to the Tip coordinate through the analytic identity $\Delta P = 0.5 + 0.25\,X$ (`test_B9`, `test_A5`). The record stands as NON-CONFORM with (i) quantified, bounded effect, (ii) a one-line correction path (a constants change, or a manual note — maintainer's decision), and (iii) no incidence on any published result (§8).

### 5.2 Polar targets of /d/ and /g/ (I6)

The engine's polar targets for the isolated consonants are carried by the same `_CONSONANT_ENTRIES` rows: /d/ at $(\rho, \theta) = (1.2,\ 23\pi/16)$, /g/ velar at $(1.2,\ \pi/3)$, /g/ palatal at $(1.1,\ 23\pi/12)$

Table 2: Register constants (Axis A) and their adjudication. Tolerances are exact identity for quoted integers/rationals of $\pi$.

| Constant (value at `4a0d76290aee`) | Article | Adjudication |
|---|---|---|
| CO rows 0,1,2,4,5,6 — e.g. Body $[-2.5, 0.0, -\pi/3]$ | Table 1 | **CONFORM**(`test_A1[0,1,2,4,5,6]`, six CONFORME verdicts; row 1 CONFORME with $\Psi_2$ identical modulo $2\pi$) |
| CO row 3 Tip: $[-2.75,\ 0.5,\ \pi]$ (`constants.py:28`) | $[-3, 0, \pi]$ | **NON-CONFORM**— $\Omega = 0.5$ undocumented in-repository (Corrigendum 1; bounded effect below) |
| `THETA_DENTAL` $= 23\pi/16$ (`:46`) | $\theta_d = 3\pi/2 = 24\pi/16$ | **DOCUMENTED DEVIATION**— genuine difference $\pi/16$, *not* modulo-$2\pi$ (verified: $\lvert 23\pi/16 - 24\pi/16\rvert = \pi/16 = 0.19635$ rad) |
| Vowel radii 0.9/0.8/0.7/0.3 (`:74-81`) | not tabulated | **DOCUMENTED DEVIATION**— documented register choice (`RHO_HI`, `RHO_MID` etc.) |
| $\rho_b = 1.2$, $\rho$ stops $= 1.2$ (`:74`) | $\rho_b$: §3 demo | **DOCUMENTED DEVIATION**— documented at `manual.tex:1076-1080, 1149-1167` |
| `VALRECT` $= 1.10$ (`config.py:43-46`) | none | **DOCUMENTED DEVIATION**— dated author ruling 2026-10-07, in-code; historical value 0.75 recorded in the same comment; effect quantified by `test_G4` (ÉCART DOCUMENTÉ, this run) |

(`phonology.py:173-180`; the palatal/velar split is exercised by `run_locus_trough_demos.py:206-216`, which restores the article positions $(1.2, 3\pi/2)$, $(1.2, \pi/3)$ and $(1.1, 23\pi/12)$). The article values are $\theta_d = 3\pi/2$, $\theta_{gv} = \pi/3$, $\theta_{gp} = 23\pi/12$ (§3, quoted in §2). The verdict structure is exactly the one the standard prescribes for such comparisons:

- /g/: both targets identical to the article ($\pi/3$ and $23\pi/12$) — conformity, no reserve needed.
- /d/: the engine's $23\pi/16$ differs from the article's $3\pi/2$ by $\pi/16$. This is a *genuine* numerical deviation: the circular distance is $\pi/16 = 0.19635$ rad, not zero modulo $2\pi$ (executed check of October 15, 2026, quoted in Annex C note). The modulo-$2\pi$ vocabulary must never be stretched to cover it.
- The genuine modulo-$2\pi$ cases are in the *lineage*, not against the article: [EXT §5] reports /d/ $-4.5\pi/8$ (Syllable_Synthesis) vs $23\pi/16$ (Timit-to-Maeda) and /g/ palatal $-\pi/12$ vs $23\pi/12$; both pairs were re-verified identical modulo $2\pi$ by execution this run ($-4.5\pi/8 + 2\pi = 23\pi/16$; $-\pi/12 + 2\pi = 23\pi/12$). The article's $3\pi/2$ differs from *both* lineage values by $\pi/16$.

Item I6 verdict, both axes: **DOCUMENTED DEVIATION** — the /d/ angle deviation is real, quantified ($\pi/16$), and documented repository-side through the differential supplement's register rows (row 1/1b and the polar-target row 4a) [6]; the repository's own demo script restores the article values where the article's demonstrations require them (`run_locus_trough_demos.py:206-216`), which is itself documentary evidence of the difference.

## 6 Trajectory mechanisms (item I7): word-onset anchoring

Difference 5 is the word-onset anchoring rule: for a non-initial word, the onset anchors on the *previous* word's `Ve` point, Vo $= (\delta_o\, \rho_{V2}, \theta_{V2})$ (`gesture.py:385-432`; docstring ruling dated 2026-

10-06 at `:399-406`). The baseline report [5] established three results, all re-confirmed by this run's executed evidence:

1. **Exact zero under published conditions.** The rule produces bitwise identical trajectories to the article's literal reading for all six published conditions ("bi bi" $\delta_o = \delta_e \in \{0.5, 0.7, 1.0\}$; "big.bi" $\delta_o = \delta_e \in \{0.7, 1.0\}$; "bi.gbi" $\delta_o = \delta_e = 1.0$): $\max|\Delta P| = 0.0$ on every articulator (artifact `d5_results.json`, re-read this run). The executed suite agrees: `test_D1-D5, D7` all CONFORME.
2. **Control divergence outside those conditions.** For "big ba" with $\delta_o = 0.5 \neq \delta_e = 0.7$ the rule diverges from the literal reading by 2.08121765715786 M.u. (Body, $t = 129$) — the largest divergence found in the audit, occurring only outside the published conditions.
3. **Structural argument.** For "big.bi" and "bi.gbi" no divergent site exists at all: the word boundary follows a consonant, so the anchoring sites are structurally absent — a proof stronger than any sampled check (`test_D6`, ÉCART DOCUMENTÉ with the rule's documentation).

Item I7 verdict (Axis A): **CONFORM** under the published conditions — the rule's effect on every article-cited demonstration is exactly zero, bitwise — with the deviation from the literal reading documented (dated in-code ruling; supplement row 5) and bounded (2.0812 M.u. on a control utterance outside the published conditions). The continuity prediction at the anchors is exact (`test_D7`: CONFORME on all four parameterizations, this run).

**Residual statement — trajectory mechanisms**

Verified: the Vo/Ve anchor formulas, the fusion condition ($\delta_o = \delta_e$, same vowel; `test_F2` CONFORME), the exact-zero result, the control divergence and its localization (Body, $t = 129$), and the structural absence of divergent sites for the article's own demonstrations. Not covered: utterances with $\delta_o \neq \delta_e$ beyond the one control, and inter-speaker or prosodic variants, for which no article reference exists.

# 7 Numerical reproduction (items I8–I9)

## 7.1 The re-executed suite and its tally

The full compliance suite `tests/compliance/` (seven suites A–G, 104 tests) was re-executed on October 15, 2026 in one session against the pinned commit: **104 executed, 104 passed**, wall time 602 s (test-time 597.5 s). The compliance tally of the fresh record is **92 CONFORM** (91 CONFORME plus one CONFORME with $\Psi_2$ identical modulo $2\pi$), **10 ÉCART DOCUMENTÉ**, **1 NON CONFORME** (CO Tip row), **1 NOT APPLICABLE** (CCC) — per-test statuses identical to the October 2026 baseline on all 104 rows (mechanical comparison of the two JSON records; zero differences). The ten documented deviations are: `D6`, `D8` (anchoring rule and extra anchors), `B9` (Tip effect on arcs), `C4[pair5-pair8]` (the four diverging pairs), `A5` (Tip sensitivity), `G2` (slope approximations), `G4` (VALRECT effect).

## 7.2 Key bounds, fresh versus baseline

Every key bound of the baseline was reproduced at the same value (the computations are deterministic and seed-free):

Table 3: Tally of the re-executed run by suite (October 15, 2026) against the baseline.

| Suite | Tests | Compliance statuses | | | | Baseline | Delta |
|---|---|---|---|---|---|---|---|
| | | CF | ED | NC | NA | | |
| A coordination | 14 | 12 | 1 | 1 | 0 | identical | none |
| B arcs | 26 | 25 | 1 | 0 | 0 | identical | none |
| C selection | 25 | 19 | 4 | 0 | 1 | identical | none |
| D anchors | 17 | 15 | 2 | 0 | 0 | identical | none |
| E graphs | 8 | 8 | 0 | 0 | 0 | identical | none |
| F transformation | 7 | 7 | 0 | 0 | 0 | identical | none |
| G locus | 7 | 6 | 1 | 0 | 0 | identical | none |
| Total | **104** | **92** | **10** | **1** | **1** | identical | none |

Table 4: Key numerical bounds: this run vs. baseline [5], with tolerances.

| Bound | This run | Baseline | Tolerance | Evidence |
|---|---|---|---|---|
| Eq. (1) internal identity | exact (0.0) | $\leq 10^{-15}$ | $5 \times 10^{-15}$ | `test_A6` |
| Arc construction vs. independent complex reference (Eq. 2/3 layer) | $\leq 2.665 \times 10^{-15}$ | $\leq 10^{-15}$–$10^{-14}$ | $5 \times 10^{-15}$ | `test_B1[14]` |
| Eq. (4) end-to-end (C3, four utterances) | $\leq 2.665 \times 10^{-15}$ | $4.4 \times 10^{-15}$ | $5 \times 10^{-15}$ | `test_C3[bi, di, gi, bib]` |
| big.bi reconstruction from article equations (F4/F4b) | $4.441 \times 10^{-15}$ (6 params); tip identity $4.052 \times 10^{-15}$ | $4.4 \times 10^{-15}$ | $5 \times 10^{-15}$ | `test_F4`, `test_F4b` |
| Cluster /gb/ end-to-end (C.4b) | $2.220 \times 10^{-15}$ (6 params); tip identity $1.166 \times 10^{-15}$ | $2.2 \times 10^{-15}$ | $5 \times 10^{-15}$ | `test_C4b` |
| Locus slopes (b, d, g-velar, g-palatal) | worst $\lvert\Delta\rvert = 4.59 \times 10^{-4}$ (g-palatal: 0.630459 vs 0.630) | $\leq 5 \times 10^{-4}$ | $5 \times 10^{-4}$ | `test_G1[4]` |
| Difference-5 effect, published conditions | 0.0 bitwise | 0.0 bitwise | exact | `d5_results.json`; `test_D1-D7` |
| Difference-5 control divergence | 2.0812 M.u. (Body, $t = 129$; full digits S11) | 2.0812 | reported | `d5_results.json` |

The transformation `big.bi` → `bi.gbi` is reconstructed by the article's equations alone — `test_F4/F4b` drive the reconstruction from Eqs. (1)–(4) with no hidden mechanism, and the residual on the six conforming parameters is $4.441 \times 10^{-15}$ (the Tip coordinate follows its own documented display convention, exact against its analytic identity). Item I8 verdict: **CONFORM** (all equation identities within the stated tolerance $5 \times 10^{-15}$). Item I9 verdict: **CONFORM** (slopes within $5 \times 10^{-4}$; `test_G2`'s ÉCART DOCUMENTÉ concerns the article's rounded two-digit approximations, e.g. 2.335 vs 2.3352211, not the equations).

# 8 Documentation conformity (items I10–I11)

## 8.1 Corrigenda status

The differential supplement [6] carries three corrigenda against the repository's manual and generator outputs; their anchors were re-opened at the pinned commit in this run and are all still accurate:

- **Corrigendum 1 (CO claim).** `manual.tex:225-227` states "the only difference is the Tip parameter $c_0$"; the same row's $\Omega = 0.5$ is omitted (`constants.py:28`). Basis of the I5 NON-CONFORM record.
- **Corrigendum 2 (Demo 4 clusters).** `manual.tex:1260-1273` (the /dg/ row at `:1268`, /gd/ at `:1270`) and the rule paragraph `:1275-1280` attribute the article's /b/-conditional rule to the engine, citing a code vector `{0,1,2,3}` that does not exist in `ART_CLUSTER` (§3).
- **Corrigendum 3 (figure labels).** `CLUSTER_INFO` (`run_article_demos.py:91-98`) hard-codes the article labels for /dg/ and /gd/ ($S_c = \{1, 2, 3, 4\}$) while the synthesis ran with $\{1, 2, 3\}$; the label flows through the console (`:310`) and the panel titles (`:356, :416`) into the embedded figures (`manual.tex:1284, :1291`). Trajectories are faithful; only labels are wrong.

## 8.2 Scope of impact

The scope-of-impact statement is reproduced from the supplement (rows 4a/4b) and re-verified by execution: the intersection of the diverging pairs {dd, dg, gd, gg} with the article's cited results is *empty* — the article's isolated consonants are conforming (§2), its only cluster is /gb/ (conforming, validated end-to-end at $\leq 2.220 \times 10^{-15}$), and its transformation demo uses /b/ and /g/ as singleton consonants. The affected utterances are confined to the extended CC simulations of the supplement's Demo 5: `adga` (`run_article_demos.py:301`) and `agda` (`:303`), two of the six CVCCV panels, with label corrections per Corrigendum 3.

## 8.3 Verdicts

Item I10 (published artifacts): **CONFORM** after Corrigenda 2–3 — trajectories and numerical outputs are faithful (suites B, E, F all conforming this run; the artifact comparison `test_B8` CONFORME); the labels of two Demo 5 panels carry the documented correction path. Item I11 (manual and supplement): **CONFORM** after Corrigenda 1–3, with the residual documentation items tracked in the supplement (the $\theta_{dental}$ note and the provenance statement), none of which affects the engine or the published numbers.

## 9 Summary conformity table

Table 5 is the certificate's contract with its reader (rule S5): one row per item, verdict per axis, and the evidence pointer (executed test or probe, plus `file:line`). Verdict vocabulary is the fixed one of the macros; "mixed" in item I5 means the row-level detail of §5, which carries one NON-CONFORM record (Tip) among conforming and documented-deviation rows.

Table 5: Item inventory and verdicts (October 15, 2026 run; commit `4a0d76290aee`).

| ID | Item | Axis | Verdict | Evidence |
|---|---|---|---|---|
| I1 | Isolated selectors /b/, /d/, /g/ | A+L | **CONFORM** | `test_C1[b,d,g]`, `test_C2`; probe; `phonology.py:173-180`; [EXT §2.3] |
| I2 | Binary-cluster masks `ART_CLUSTER` | L | **CONFORM** | `constants.py:89-92`; [EXT §3.1] |
| | (same masks vs. article rule) | A | **DOCUMENTED DEVIATION** | probe 4/9 diverge; `test_C4[pair5, 6, 7, 8]`; Corr. 2 |
| I3 | Cluster index rule (threshold) | L | **CONFORM** | `phonology.py:337-341`; [EXT §3.2] |
| I4 | Cluster activation, $m = 3$ branch | L | **CONFORM** | `phonology.py:345-352`; [EXT §4.3]; `test_E7` |
| | (article reference for CCC) | A | **NOT APPLICABLE** | `test_C5` (reason recorded) |
| I5 | Register constants (detail: Table 2) | A | mixed: see §5 | Table 2; `test_A1`, `A5`, `G4` |
| I6 | Polar targets /d/, /g/ | A+L | **DOCUMENTED DEVIATION** | $\pi/16$ executed check; [EXT §5]; `run_locus_trough_demos.py:206-216` |
| I7 | Word-onset anchoring (difference 5) | A | **CONFORM** (published conditions) | 0.0 bitwise; control 2.0812; `test_D1-D8`; `gesture.py:385-432` |
| I8 | Article Eqs. (1)–(4), end-to-end | A | **CONFORM** ($\leq 5 \times 10^{-15}$) | `test_A6`, `B1`, `C3`, `F4/F4b`; Table 4 |
| I9 | Locus equations and slopes | A | **CONFORM** ($\leq 5 \times 10^{-4}$) | `test_G1[4]`: worst $4.59 \times 10^{-4}$ |
| I10 | Published artifacts (demos, figures) | A | **CONFORM** after Corr. 2–3 | `test_B8`, suites E/F; §8 |
| I11 | Documentation (manual, supplement) | A | **CONFORM** after Corr. 1–3 | [6]; §8 |
| I12 | Environment and reproducibility | — | **CONFORM** | §1.2; fresh `test_results.json`; tally identical 104/104 |

Reading the table mechanically per §5.2 of the brief: no NON-CONFORM item exists beyond the recorded Tip row; every equation identity is within tolerance; the scope-of-impact statement is reproduced and still true; every DOCUMENTED DEVIATION carries its provenance citation. The overall verdict follows (§11).

## 10 Verification checklist

The following numbered steps re-derive every identity claim of this certificate from the repository alone (rule S6); steps 1–7 transpose the standard's checklist [EXT §8] to the target, steps 8–10 are the executed additions of this run. Each step's expected output was written before execution, and each was executed on October 15, 2026 at the pinned commit.

1. Read `synthSYL/constants.py:89-92`: `ART_CLUSTER` must be exactly `[[0,1,2,5],[0,1,2]]`. Expected: two masks, four entries and three entries.

2. Read `synthSYL/phonology.py:133-141` and `:337-341`: place coding b= 0, d= 1, g= 2; rule `int(idx_c1 + idx_c2 > 2)`. Expected: threshold rule selecting mask row 1 when the place-index sum exceeds 2.

3. Read `synthSYL/phonology.py:173-180`: isolated entries with selectors `[0,1,5]` for /b/ and `[0,1,2,3]` for /d/, /g/; convert by +1. Expected: $\{1, 2, 6\}$ and $\{1, 2, 3, 4\}$.

4. Read `synthSYL/phonology.py:345-352`: the $m = 3$ branch must be present with the generic $\{0, 1, 2\}$ construction. Expected: branch present (`params.add(5)` for labial, `params.add(6)` for uvular).

5. Run the nine-pair probe (Annex C). Expected output: five `conform` lines (bb, bd, bg, db, gb) and four `DIVERGES` lines (dd $\to \{1, 2, 3, 6\}$; dg, gd, gg $\to \{1, 2, 3\}$).

6. Read `synthSYL/constants.py:28` and `docs/manual.tex:225-227`: Tip row $[-2.75, 0.5, \pi]$ against the manual's "only difference is $c_0$" claim. Expected: the $\Omega = 0.5$ omission (Corrigendum 1) is visible.

7. Compare polar targets: `constants.py:46` ($23\pi/16$) against the article's $3\pi/2$; verify the difference is $\pi/16$ and *not* zero modulo $2\pi$; separately verify $-4.5\pi/8 \equiv 23\pi/16$ and $-\pi/12 \equiv 23\pi/12$ (modulo $2\pi$) for the lineage pair [EXT §5].

8. Execute `pytest tests/compliance/` (104 tests) with the pinned venv. Expected: 104 passed; tally 92/10/1/1; statuses identical to `test_results.json` of October 15, 2026.

9. Re-derive the key bounds of Table 4 from the fresh JSON record. Expected: Eq. (4) layer $\leq 4.441 \times 10^{-15}$; /gb/ $\leq 2.220 \times 10^{-15}$; slope worst $|\Delta| \leq 4.6 \times 10^{-4}$; D5 exact zero; control 2.0812 M.u.

10. Confirm the engine is untouched: `git diff -stat` over `synthSYL/`, `vlam.py`, `config.py` shows zero content change (mode bits only), and the certificate's sources live outside those paths.

## 11 Attestation statement

The mechanical decision rule of the brief has been applied to the tally of §7 and the item verdicts of §9: (i) no NON-CONFORM item exists beyond the recorded Tip row; (ii) every equation identity is within the stated tolerance ($\leq 5 \times 10^{-15}$; worst observed $4.441 \times 10^{-15}$); (iii) the scope-of-impact statement is reproduced and still true (no article-cited result affected); (iv) every DOCUMENTED DEVIATION carries its provenance citation. The overall verdict is therefore:

Attestation statement

**Overall verdict: CONFORM WITH DOCUMENTED DEVIATIONS**

**Target:** repository `big.bi-supplement`, commit `4a0d76290aee` (DOI 10.5281/zenodo.23239483), verified October 15, 2026.

**Axes:** Axis L (lineage identity to the 2023 engines) and Axis A (conformity to arXiv:2307.02299), reported separately throughout.

**Tally:** 104 tests executed, 104 passed in one session (602 s wall); 92 CONFORM / 10 DOCUMENTED DEVIATION / 1 NON-CONFORM / 1 NOT APPLICABLE; per-test statuses identical to the October 2026 baseline (104/104 rows).

**Key bounds:** Eq. (4) end-to-end $\leq 4.441 \times 10^{-15}$; /gb/ end-to-end $\leq 2.220 \times 10^{-15}$; slopes within $4.59 \times 10^{-4}$; difference-5 exactly zero (bitwise) under the six published conditions; control divergence 2.08121765715786 M.u. (Body, $t = 129$).

**NON-CONFORM record:** CO matrix Tip row (`constants.py:28`), $\Omega = 0.5$ undocumented (Corrigendum 1); bounded effect (counterfactual substitution restores identity at $\leq 1 \times 10^{-15}$ on all seven checks); correction path: one-line constants change or manual note — maintainer's decision.

**Attesting agent:** Super Z (GLM, Z.ai), automated verification run; every verdict in this certificate rests on evidence executed or re-opened during the October 15, 2026 run recorded in Annex A.

**Limitations:** this attestation covers the selection layer, register constants, trajectory mechanisms, numerical reproduction of the article's equations and announced results, and the documentation state after Corrigenda 1–3, all at the pinned commit. It does not extend to acoustic recordings, perceptual judgments, three-consonant clusters (no article reference), or any future change to the repository; any engine change voids the pin and requires full re-execution (§1.3).

Signature — attesting agent, October 15, 2026. Certificate ID: BBICERT-2026-10-15-4a0d76290aee.

# A Annex A — Verification record and quote register

## A.1 Citations re-opened in this run

Every `file:line` citation of this certificate was re-opened and read at the pinned commit on October 15, 2026, in the same session that executed the suite (QA rule of the brief). The record, by file:

- `constants.py:6-8`, `:24-32` (CO matrix; Tip row `:28`), `:46` (`THETA_DENTAL`), `:74-81` (radii), `:89-92` (`ART_CLUSTER`);
- `phonology.py:133-141` (`_TABCONS_IDX`), `:173-180` (isolated entries), `:303-304` (singleton fallback), `:337-341` (threshold rule), `:345-352` ($m = 3$ branch — new citation established this run);
- `gesture.py:385`, `:399-406` (docstring ruling); `config.py:43-46` (`VALRECT`);
- `run_locus_trough_demos.py:156`, `:206-216` (article-position restoration);

- `run_article_demos.py:81`, `:91-98` (`CLUSTER_INFO`), `:296-304` (`adga` at `:301`, `agda` at `:303`), `:310`, `:356`, `:416`;
- `manual.tex:225-227`, `:1076-1080`, `:1149-1152`, `:1165-1167` ($\rho_b$ documentation), `:1260-1273` (rows `:1268`, `:1270`), `:1275-1280`, `:1284`, `:1291`.

Environment facts: Python 3.12.14 / numpy 2.5.3 / scipy 1.18.1 / matplotlib 3.11.2 / pytest 9.1.1 confirmed by the interpreter and recorded in the fresh `test_results.json` metadata.

### A.2 Article quotes used (verbatim, arXiv:2307.02299)

- §3, isolated: "... For the /g/ this is well known that it is velar for back vowels and palatal for front vowels. This led to the locations ($\rho_{gv} = 1.2, \theta_{gv} = \pi/3$) and ($\rho_{gp} = 1.1, \theta_{gp} = 23\pi/12$) combined with the selection vector $S_c = \{1, 2, 3, 4\}$ involving tongue and jaw. The location of /d/ depends on the following vowel around ($\rho_d = 1.2, \theta_d = 3\pi/2$) and /d/ has same selection vector $S_c = \{1, 2, 3, 4\}$."
- §3, clusters: "The tuning of the six pair clusters of /b,d,g/ follows the same principle, but there is a special grouping of articulators when /b/ is involved in the pair. In this case $S_c = \{1, 2, 3, 6\}$ and otherwise this is $S_c = \{1, 2, 3, 4\}$. When involved, the /g/ is systematically palatal."

### A.3 Standard quotes used (verbatim, [EXT], October 13, 2026)

- §2.3: "The isolated selectors are identical after index conversion."
- §3.1: "The masks are therefore identical." (with the inert duplicated Jaw entry: "Double selection of the same parameter has no effect on the trajectory.")
- §3.2: the rule art_idx = $\mathbf{1}[\text{idx}_{c1} + \text{idx}_{c2} > 2]$, place coding b,p $\to$ 0; d,t $\to$ 1; g,k $\to$ 2; "The decision rule is the same."
- §5: "Selection masks can be identical while polar targets differ." — /d/: $-4.5\pi/8$ vs $23\pi/16$; /g/ palatal: $-\pi/12$ vs $23\pi/12$ (identical modulo $2\pi$).
- §9: "... the two engines implement the same selection policy. Differences in polar targets or other global constants remain possible sources of numerical divergence and must be checked separately."

## B Annex B — Item-to-test map

Map between the certificate's items and the executed suites (fresh `test_results.json`, October 15, 2026). Statuses: CF = CONFORME, ED = ÉCART DOCUMENTÉ, NC = NON CONFORME, NA = NOT APPLICABLE.

| Item | Suite | Executed tests | Statuses | Note |
|---|---|---|---|---|
| I1 | C selection | `test_C1[b,d,g]`, `test_C2[3]` | 6 CF | vectors and partition |
| I2/I3 | C selection | `test_C4[pair0-8]` | 5 CF + 4 ED | pair5–8 = dd, dg, gd, gg |
| I2 (numerical) | C selection | `test_C4b`, `test_C6` | 2 CF | /gb/ end-to-end $\leq 2.22 \times 10^{-15}$ |
| I4 | C, E | `test_C5`; `test_E7` | 1 NA + 1 CF | CCC out of article scope; branch exercised |
| I5 | A, G | `test_A1[0-6]`, `A5`, `G4` | 6 CF + 3 ED + 1 NC | A1[3] = Tip row (NC record) |
| I6 | A, C | `test_C1[d,g]` angles; probe note | ED (row) | $\pi/16$ executed check |
| I7 | D, F | `test_D1-D8`, `F1-F3` | 15 CF + 2 ED + 1 – | D6/D8 = documented rule/anchors |
| I8 | A, B, C, F | `A6`, `B1[14]`, `B2[2]`, `B3[2]`, `C3[4]`, `F4`, `F4b` | all CF | $\leq 4.441 \times 10^{-15}$ worst |
| I9 | G | `test_G1[4]`, `G2`, `G3` | 5 CF + 1 ED | G2 = rounded article approximations |
| I10 | B, E, F | `test_B8`, `E1-E8`, `F5` | 10 CF | artifact/figure layer |
| I11 | — | citation re-opening (Annex A) | — | Corrigenda 1–3 anchors re-read |
| I12 | — | environment block + JSON metadata | — | §1.2 |

# C Annex C — Probe raw output and executed fact checks

## C.1 Nine-pair selector probe (P5, October 15, 2026)

Probe source: `cluster_art_params` called on the nine place pairs, converted by +1; article rule from §3 ($\{1,2,3,6\}$ if /b/ involved, else $\{1,2,3,4\}$). Raw output (verbatim):

```
/bb/  engine [1, 2, 3, 6]  article [1, 2, 3, 6]  conform
/bd/  engine [1, 2, 3, 6]  article [1, 2, 3, 6]  conform
/bg/  engine [1, 2, 3, 6]  article [1, 2, 3, 6]  conform
/db/  engine [1, 2, 3, 6]  article [1, 2, 3, 6]  conform
/dd/  engine [1, 2, 3, 6]  article [1, 2, 3, 4]  DIVERGES
/dg/  engine [1, 2, 3]    article [1, 2, 3, 4]  DIVERGES
/gb/  engine [1, 2, 3, 6]  article [1, 2, 3, 6]  conform
/gd/  engine [1, 2, 3]    article [1, 2, 3, 4]  DIVERGES
/gg/  engine [1, 2, 3]    article [1, 2, 3, 4]  DIVERGES
```

Expected partition (written before execution): five `conform`, four `DIVERGES`. Observed: identical. Place-index sums: bb 0, bd 1, bg 2, db 1, dd 2, dg 3, gb 2, gd 3, gg 4 — threshold at sum $> 2$.

### C.2 Executed fact checks (B4/B8, October 15, 2026)

Raw output of the angular-comparison script (values at machine precision):

```
B4: 23*pi/16 - 3*pi/2 = -0.19634954084936207 rad = -pi/16
    -> genuine pi/16 difference, NOT modulo-equivalent
B8: -4.5pi/8 vs 23pi/16 : identical mod 2pi: True
    -pi/12  vs 23pi/12 : identical mod 2pi: True
    article 3pi/2 vs both lineage values: circular distance pi/16
```

## D Annex D — References


[1] `big.bi-supplement` repository, commit `4a0d76290aee`, DOI 10.5281/zenodo.23239483. Files cited: `synthSYL/constants.py`, `synthSYL/phonology.py`, `synthSYL/gesture.py`, `config.py`, `scripts/run_article_demos.py`, `scripts/run_locus_trough_demos.py`, `docs/manual.tex`.

[2] F. Berthommier, "Why can *big.bi* be changed to *bi.gbi*? A mathematical model of syllabification and articulatory synthesis," arXiv:2307.02299, Interspeech 2023.

[3] "Comparative audit of consonant selection vectors $C \in \{\mathrm{b}, \mathrm{d}, \mathrm{g}\}$ in Timit-to-Maeda and Syllable_Synthesis," external technical note, 6 pp., October 13, 2026 [EXT].

[4] Lineage comparators `Timit-to-Maeda` and `Syllable_Synthesis` (2023 engines), compared in [3] through verbatim excerpts; not re-downloaded for this run.

[5] 104-test compliance audit and difference-5 report, October 2026: `compliance_report.md`, `test_results.json`, `difference5_effect_audit.pdf`, executable suite `tests/compliance/` (suites A–G), reference equations and tolerances `reference_parameters.json`.

[6] Differential supplement (documentation-side record with Corrigenda 1–3 and scope rows 4a/4b), `differential_supplement.pdf`, October 13, 2026.


**Fresh execution record of this certificate:** `test_results.json` (October 15, 2026, 104 entries), `d5_results.json`, nine-pair probe log and angular-comparison log (Annex C); all retained alongside the certificate sources.